



Active Inference account of orofacial mimicry in **Misophonia**

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Misophonia

hatredness of bodily generated sounds



Trigger sounds

Common trigger sounds can be:

 Chewing food noisily.	 Ticking of a clock.	 Heavy breathing.
 Tapping fingers or toes.	 Clicking a pen.	 Water dripping.
 Rustling of paper or plastic.	 Smacking lips.	

Reaction

Misophonia

The symptoms of misophonia revolve around how you react to trigger sounds.

Reactions can be:

 Emotional feelings like anger.	 Body processes like a faster heartbeat.	 Behavioral actions like glaring.
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Symptoms

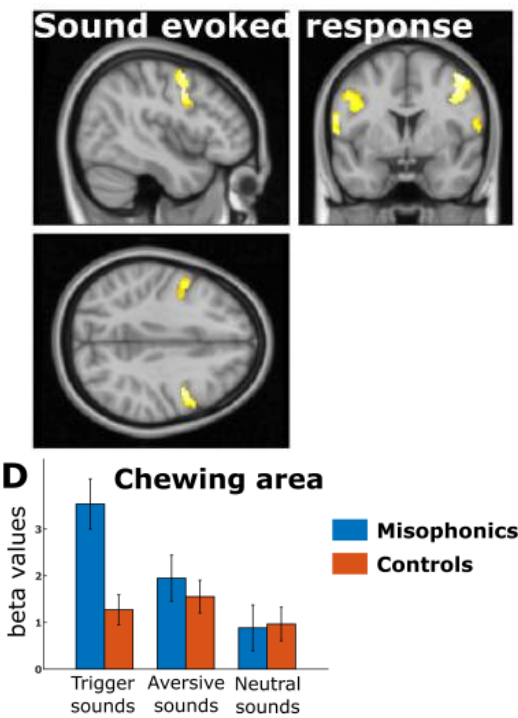
What Are the Symptoms of Misophonia?

A person with misophonia or selective sound sensitivity syndrome might experience any of the following symptoms, which can range from mild to severe:

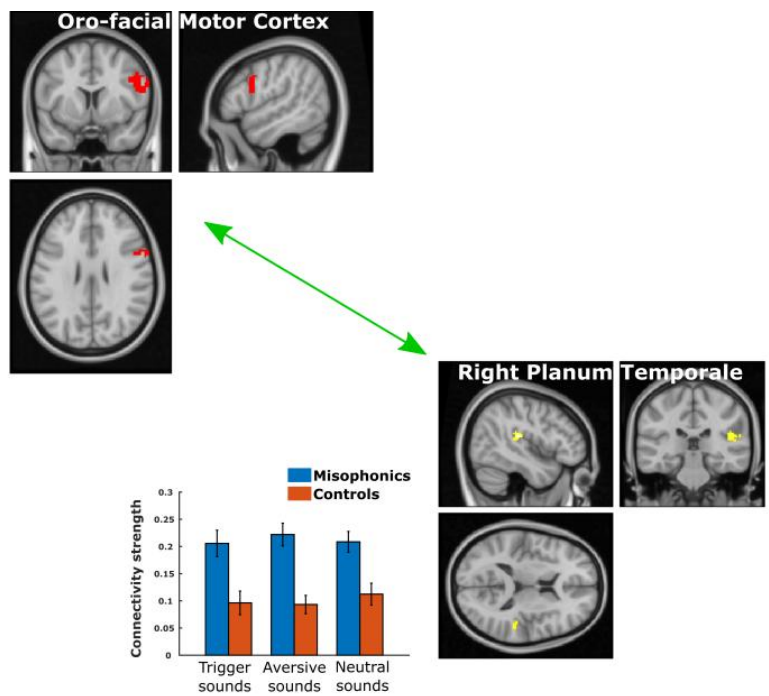
• Anxiety	• Panic
• Disgust	• Hatred
• Discomfort	• Fear
• Rage	• Emotional distress

Response to Triggers sounds

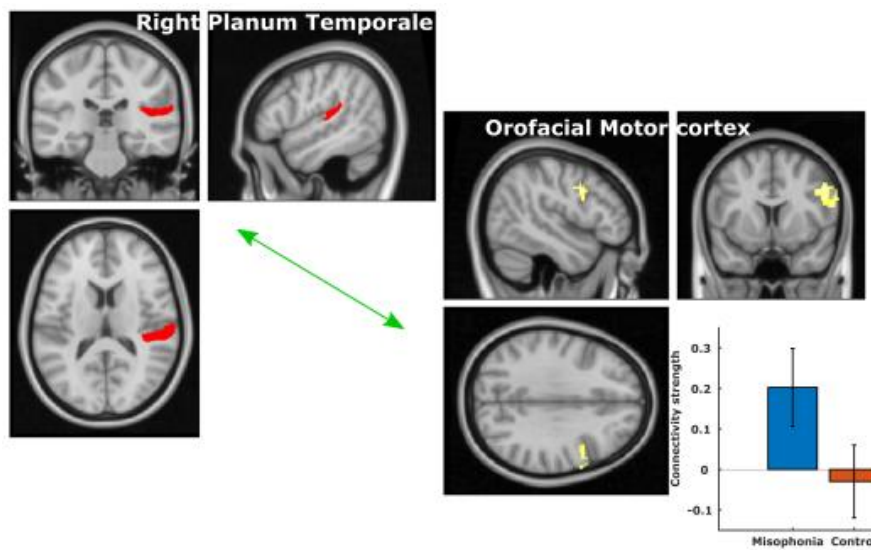
F conn. at rest



Sound Evoked Response

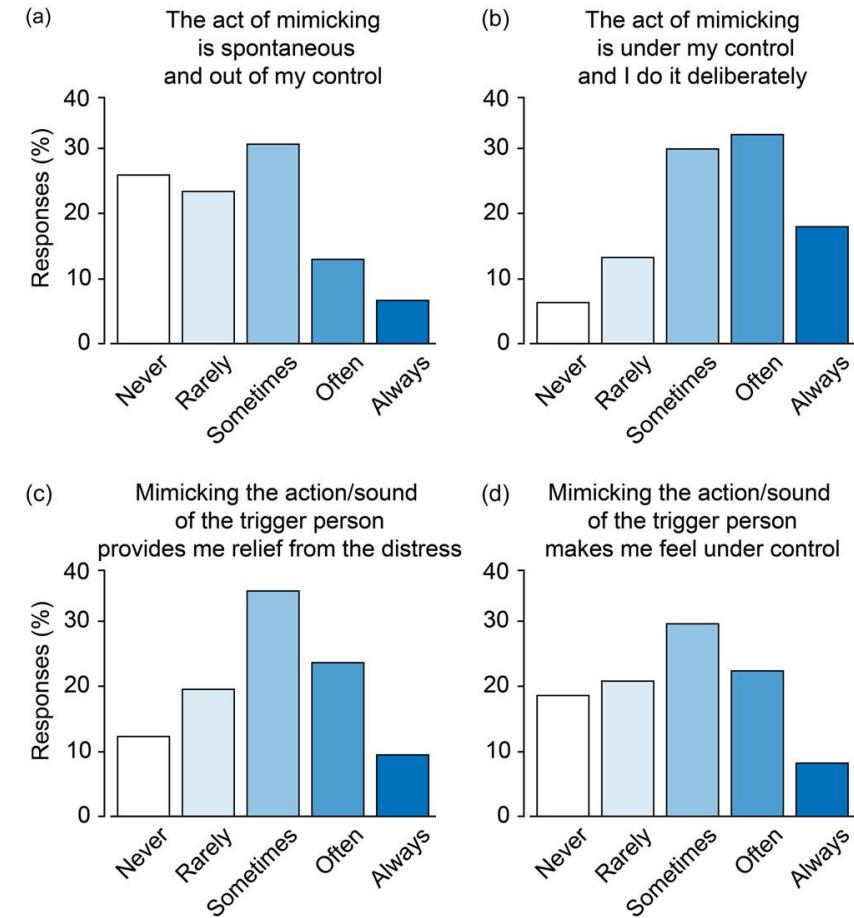
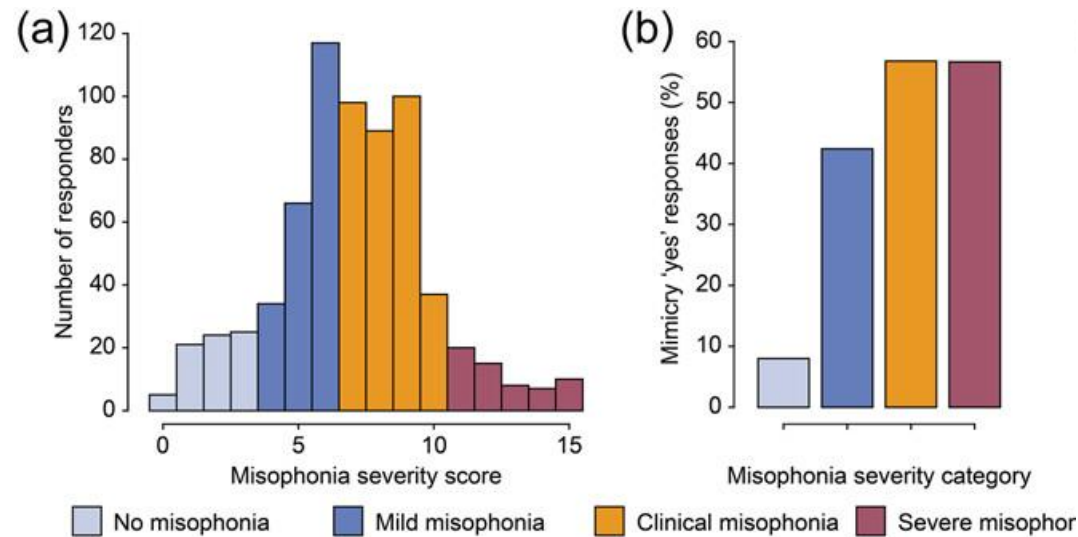


Functional Connectivity during sound perception



Resting state Functional Connectivity

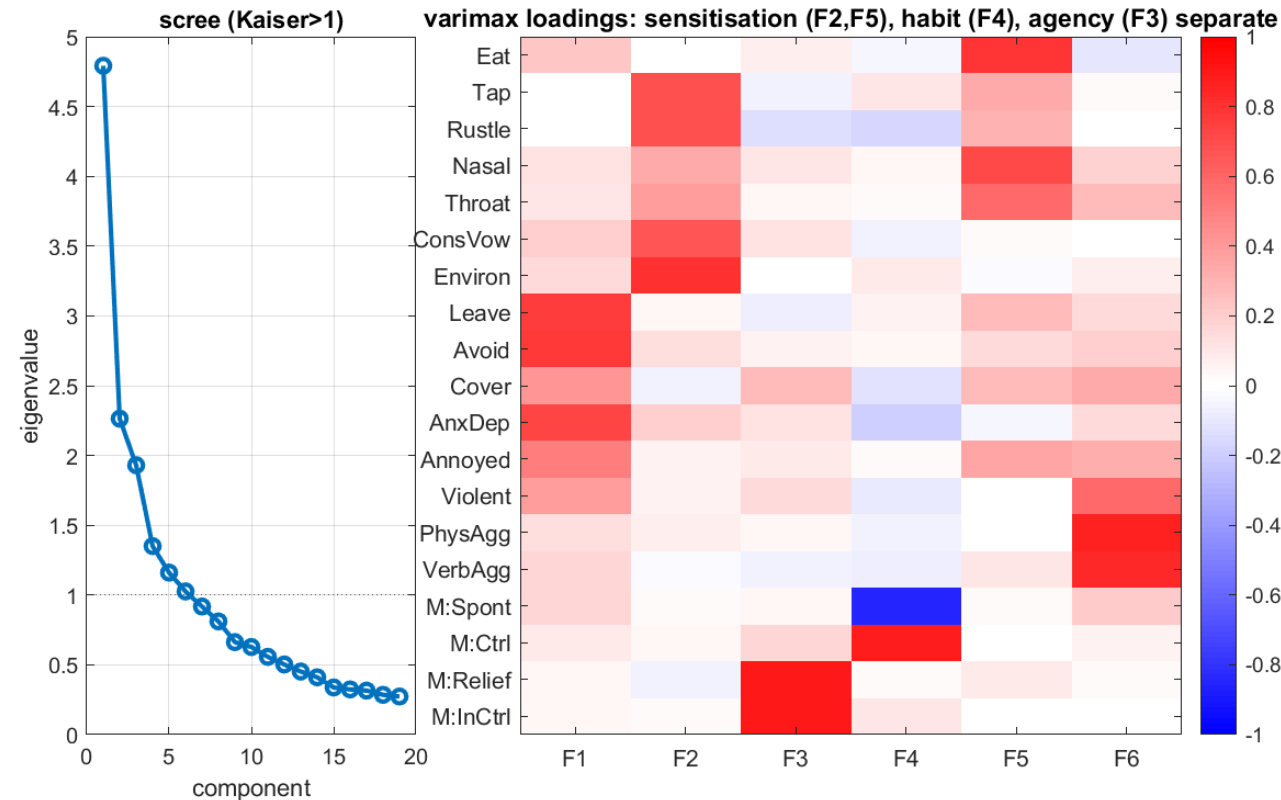
Mimicry in misophonia: A large-scale survey



Mimicry Survey – Factor Analysis

Six-factor solution (65% var)

- F1 – Avoidance + distress
- F2 – Non-orofacial triggers
- F3 – Mimicry agency / relief
- F4 – Mimicry habit / automaticity
(spontaneous –, under-control +)
- F5 – Orofacial triggers
- F6 – Aggression



Active Inference

- Puzzling - Why would mimicking the action & the annoying trigger sound help?
- Active inference provides a normative framework
 - perception, action are unified - Minimisation of variational free energy
- Tenets of active inference
 - a) Agents *act to make their predictions come true*
(action fulfils prior beliefs by changing sensation, rather than being commanded directly)
 - b) Agents must transiently attenuate the precision of the sensory consequences of their own actions if they need to act

Self-generated sensations are down-weighted (corollary discharge / sensory attenuation), so that descending motor predictions are not immediately overridden by ascending sensory evidence

Active Inference – Mimicry in Misophonia

- Misophonic mimicry is a textbook instance of Active Inference.
- Hearing an orofacial trigger engages the observer's motor model i.e. an implicit prediction "I am producing this action".
- But the observer is not the author: the sound is exogenous and socially uncontrollable.
- So the motor-resonant prediction generates a high-precision prediction error that can be nulled by neither perception (the sound keeps arriving) nor ordinary action (cannot stop another's chewing).
- That un-resolvable, precision-weighted error is the misophonic **distress**.
- Mimicry resolves it by changing the author: performing the matching orofacial act
 - a) fulfils the resonant prediction
 - b) recruits the sensory-attenuation machinery reserved for self-generated acts, so the reafferent sound becomes predicted and attenuated.
- Mimicry thus converts an uncontrollable exogenous cause into a controllable, attenuable endogenous one i.e. it restores agency.

Active Inference modelling

#	Model	Question	Key result
M1	Discrete POMDP	Is the urge to mimic an EFE-minimising policy?	Trigger→distress→mimic→relief; urge ($P(\text{mimic})$) rises with severity; orofacial-specific
M2	Emergent	Can relief emerge?	Distress/relief emerge as precision-weighted prediction error; suppression sustains distress
M3	Precision Choice	Can attenuation be an inferred (mental) action?	Agent optimises sensory precision; mimicry vs withdrawal chosen by cost; deeper relief from authorship
M4	Dual Precision	What if proprioceptive precision is also inferred?	Third route (numbing/dissociation) emerges; exteroceptive / interoceptive asymmetry
M5	ALAP (continuous)	Does attenuation transiently gate action?	Transient, self-timed attenuation emerges; self-produced sound registered $\sim 17\times$ less precisely than identical external sound
M6	Learning	How does the urge become a compulsion?	Habit (policy Dirichlet) entrenches mimicry; generalises to silence; override threshold for protective goals
M7	Vicious Cycle	Do sensitisation and habit interact?	Chronic state is a joint product of two separable channels (perceptual sensitisation $\times$ procedural habit)

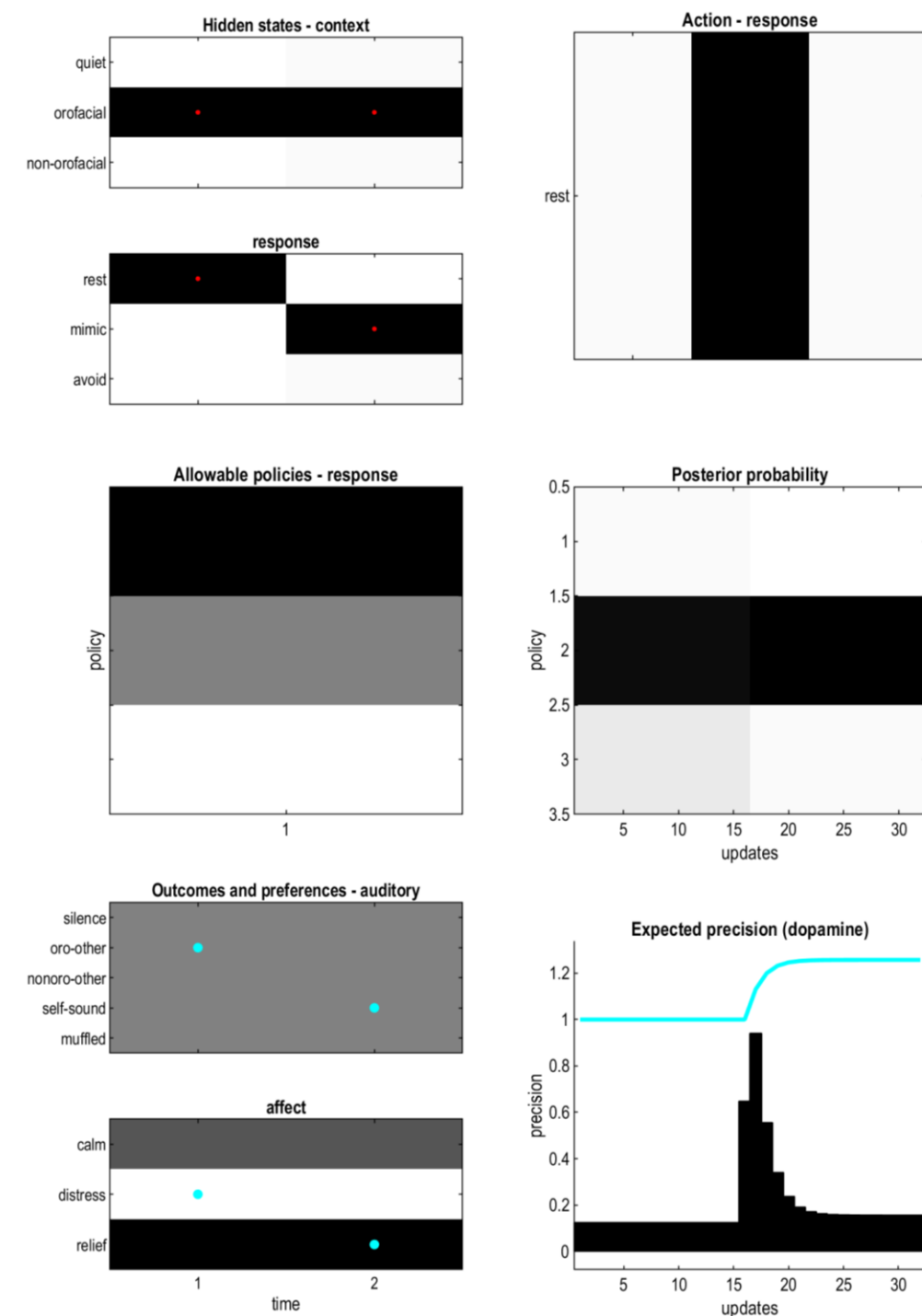
M1 – Discrete POMDP

Question:

- Is the urge to mimic an EFE-minimising policy?

Results:

- Trigger → distress → mimic → relief
- Urge ($P(\text{mimic})$) rises with severity
- Orofacial-specific



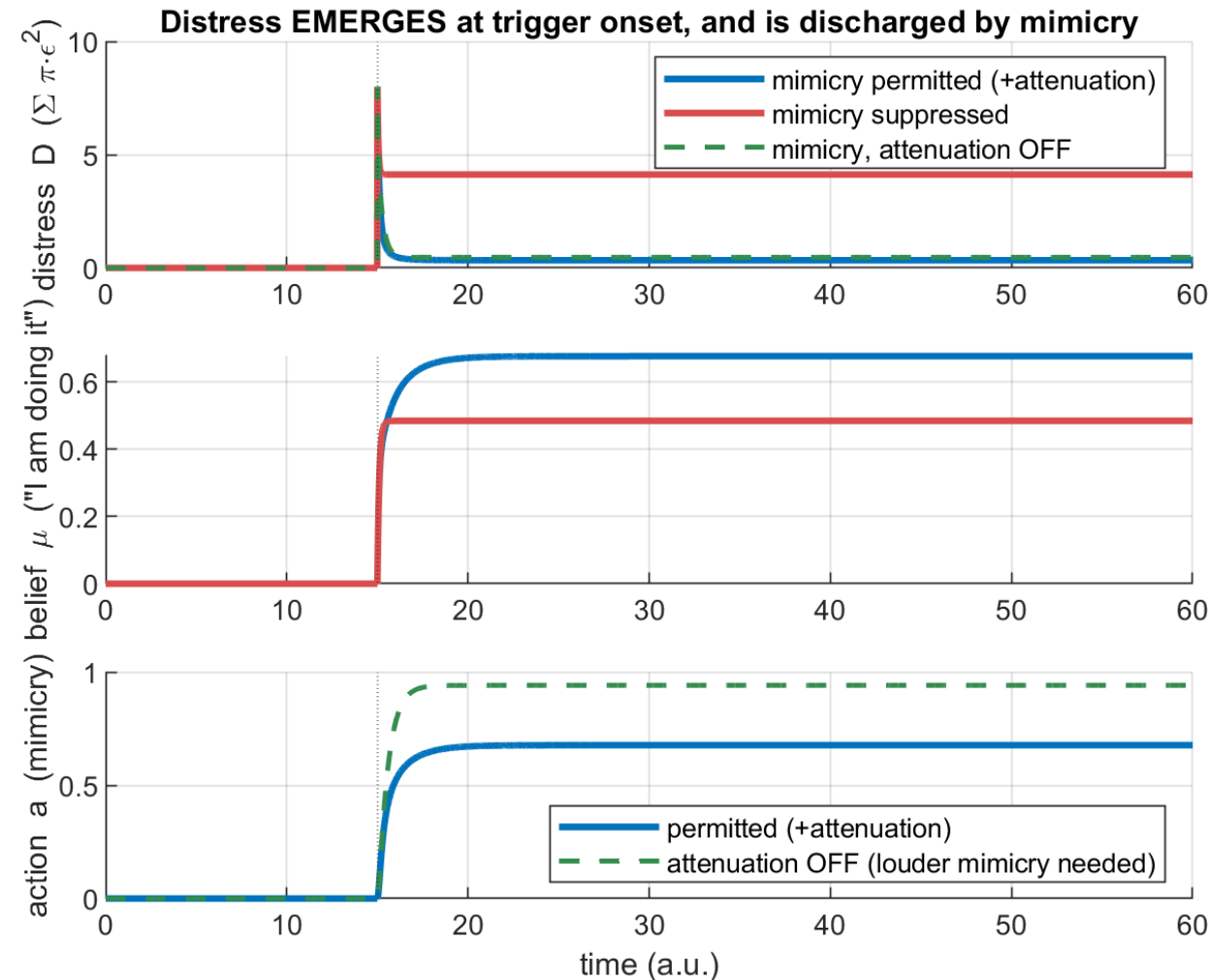
M2 – Emergent

Question:

- Can relief emerge?

Results:

- Distress / relief emerge as precision-weighted prediction error
- Suppression sustains distress



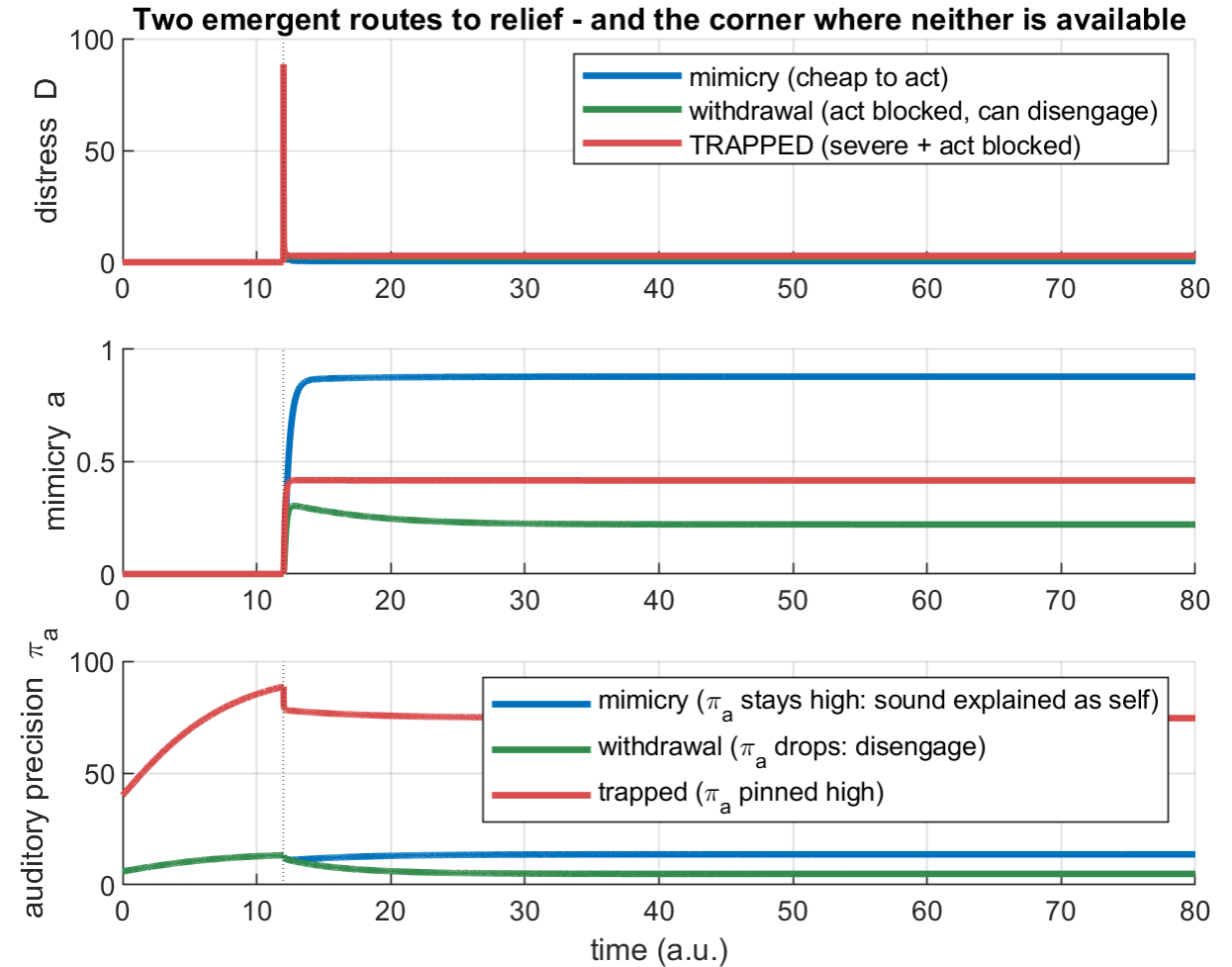
M3 – Precision Choice

Question:

- Can attenuation be an inferred (mental) action?

Results:

- Agent optimises sensory precision
- Mimicry vs withdrawal chosen by cost
- Deeper relief from authorship



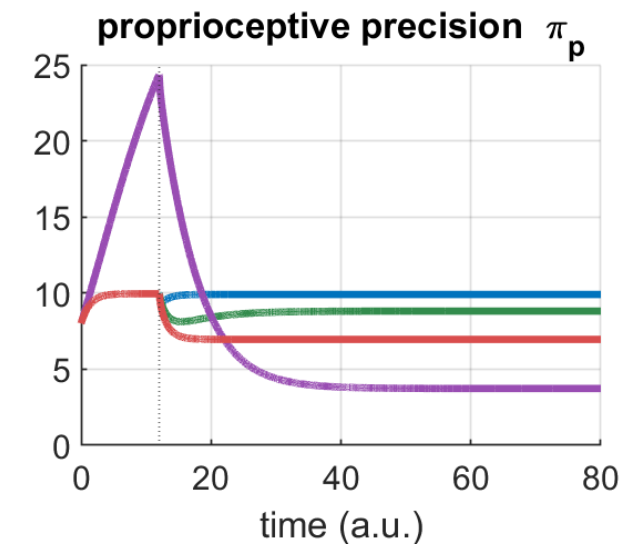
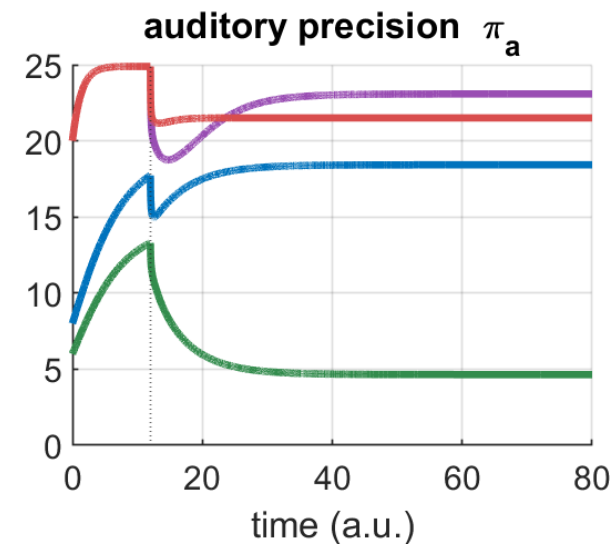
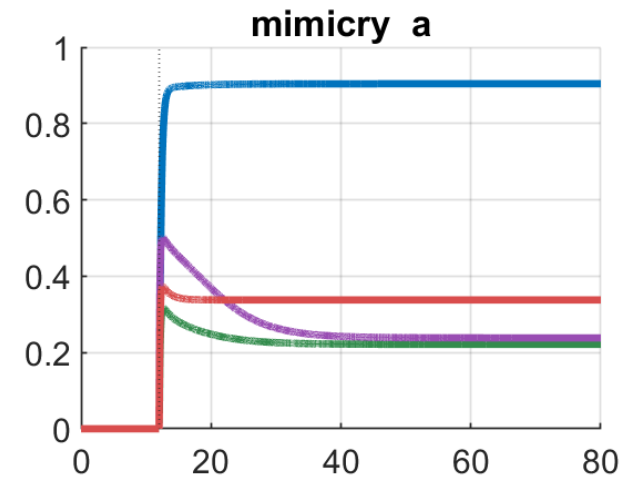
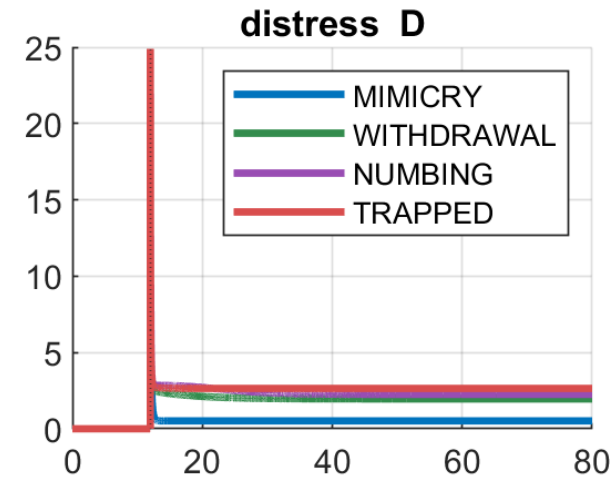
M4 – Dual Precision

Question:

- What if proprioceptive precision is also inferred?

Results:

- Third route (numbing / dissociation) emerges
- Exteroceptive / interoceptive asymmetry



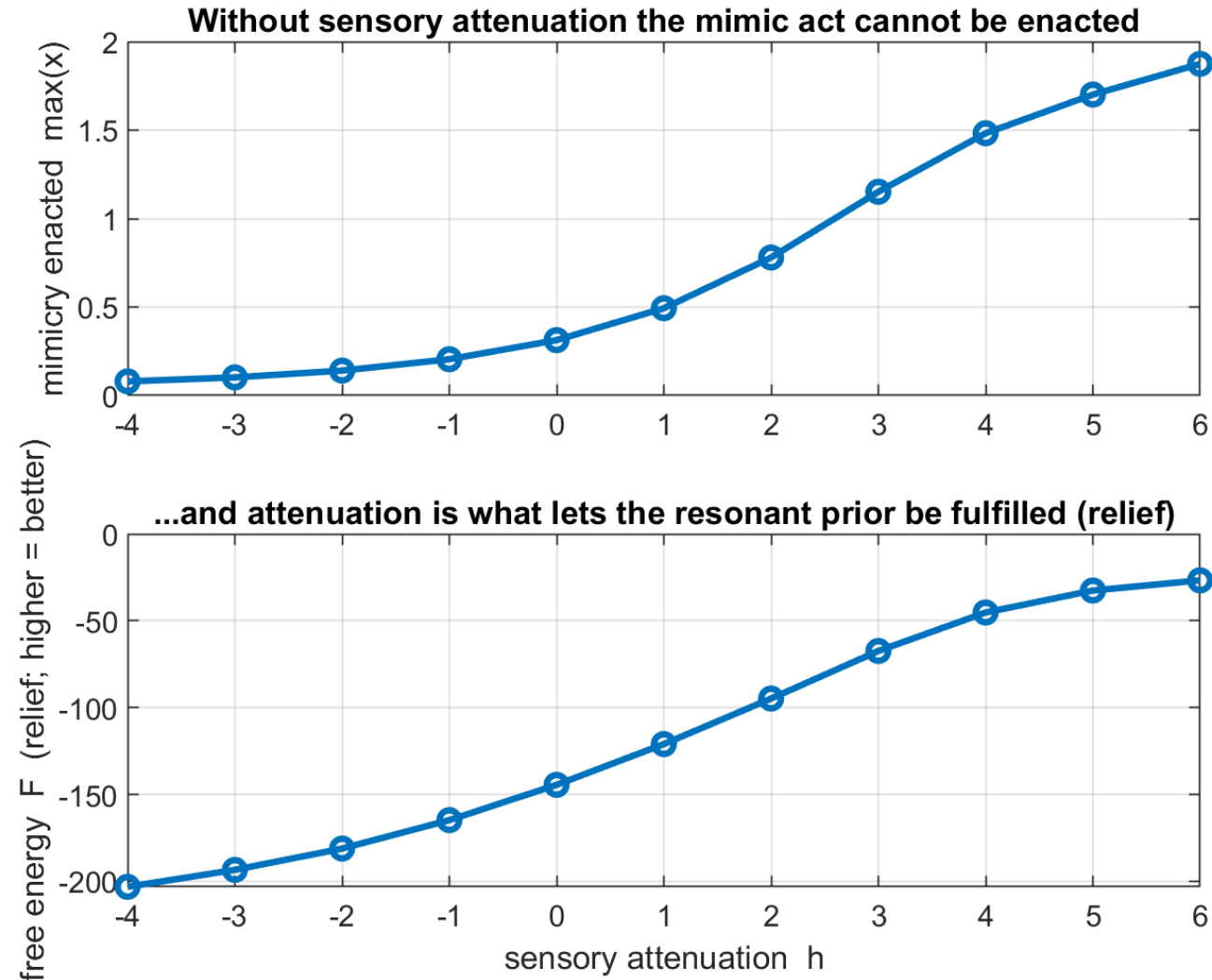
M5 – ALAP (continuous)

Question:

- Does attenuation transiently gate action?

Results:

- Transient, self-timed attenuation emerges
- Self-produced sound registered $\sim 17\times$ less precisely than identical external sound



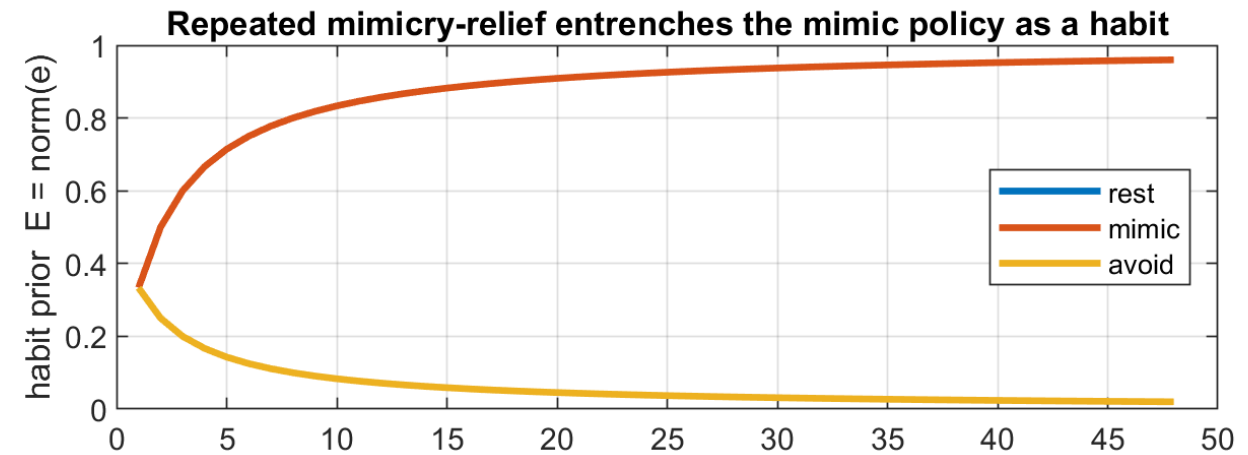
M6 – Learning

Question:

- How does the urge become a compulsion?

Results:

- Habit (policy Dirichlet) entrenches mimicry
- Generalises to silence
- Override threshold for protective goals



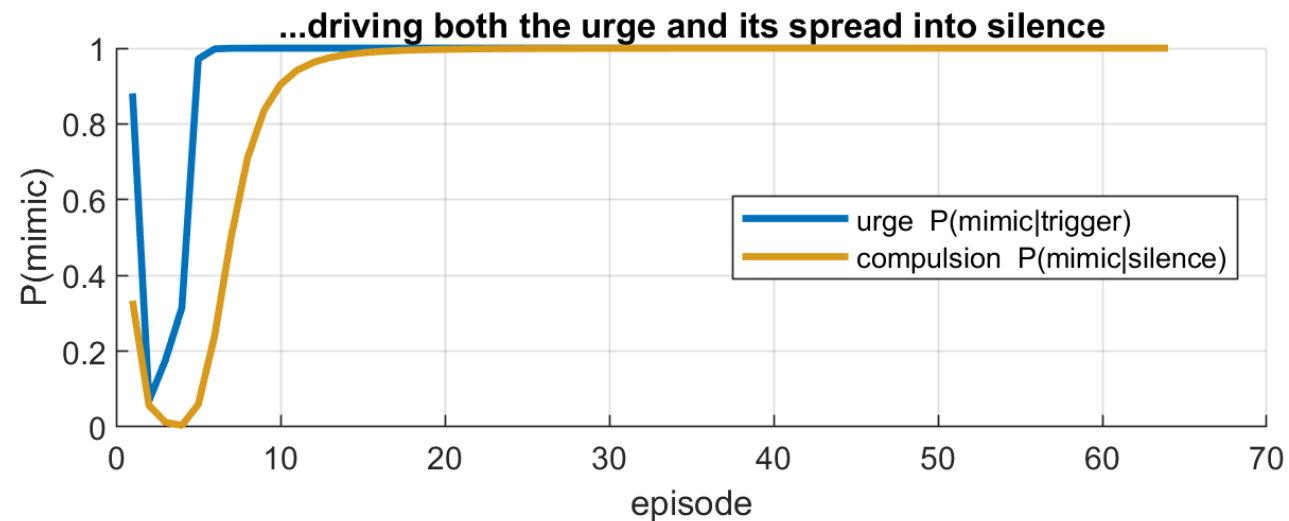
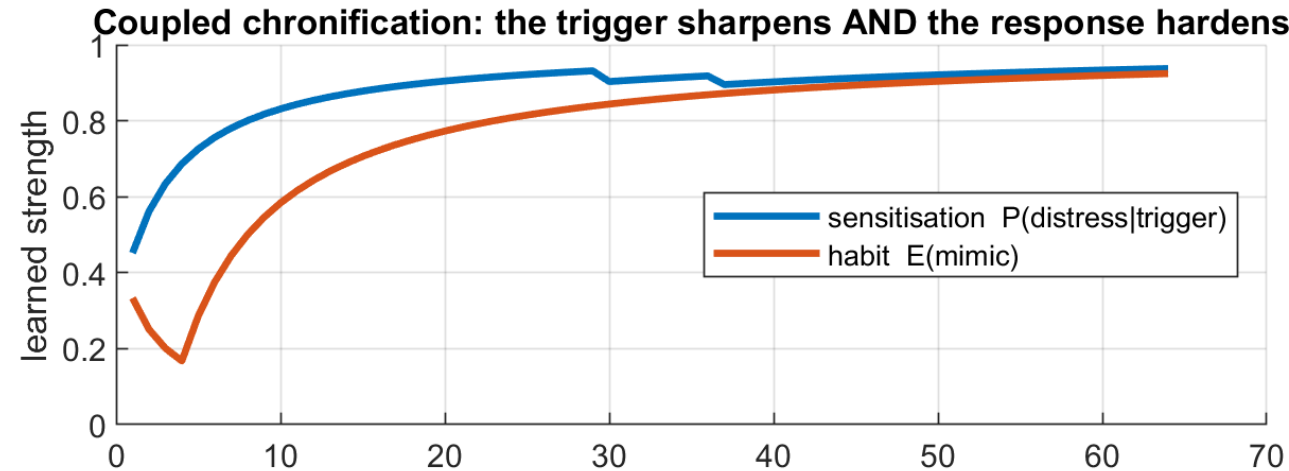
M7 – Vicious Cycle

Question:

- Do sensitisation and habit interact?

Results:

- Chronic state is a joint product of two separable channels
- perceptual sensitisation × procedural habit



Conclusions

- A small set of active-inference principles is sufficient to generate the phenomenology of orofacial mimicry in misophonia
 - Action as the fulfilment of predictions
 - Sensory attenuation of self-generated acts
 - Optimisation of precision
- Mimicry in Misophonia large scale survey support the active inference account's mechanistic and structural claims
 - Relief is agency-restoration
 - Automatic mimicry is involuntary.
- Misophonic mimicry is not a curiosity but a rational, even if costly, attempt to make an unbearable prediction come true.

References

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6. Scientific American Magazine (Sept 2021), www.scientificamerican.com/article/misophonia-might-not-be-about-hating-sounds-after-all/

